

EXAMVAULT

College Exam Resource Sharing System

PROJECT REPORT

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Specialization	Artificial Intelligence and Machine Learning
Course	Programming in Java

1. Introduction

ExamVault is a command-line Java application developed to provide college students with a centralized system for sharing and accessing examination-related resources. The system supports previous question papers, important questions, important topics, and teacher instructions. Resources are organized by subject, examination, slot group, semester, and academic year.

2. Problem Statement

College students often obtain examination resources through classmates, messaging groups, and personal collections. These resources can become scattered, making it difficult to find the correct material for a particular subject, examination, semester, or slot group. ExamVault addresses this problem by providing a centralized command-line system where students can register, upload resources, browse resources, and search using relevant filters.

3. Objectives

- Centralize college examination resources.
- Allow students to upload question papers and study resources.
- Organize resources by subject, exam, slot group, semester, and academic year.
- Provide browsing and search functionality.
- Apply Java Object-Oriented Programming concepts to a practical problem.

4. Functional Requirements

Functional Module	Description
Student Management	Registration and login using registration number and password.
Question Paper Upload	Upload a question paper with academic metadata and file path.
Important Questions	Store important questions for a subject and examination.
Important Topics	Store important topics for a subject and examination.
Teacher Instructions	Store teacher instructions for an examination.
Browse Resources	Display resources stored in the system.

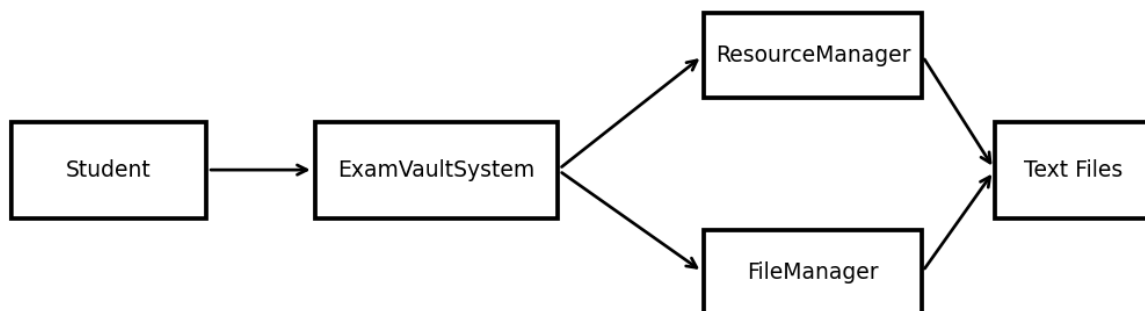
5. Non-Functional Requirements

- Usability – simple menu-driven command-line interaction.
- Reliability – validation for invalid credentials, duplicate registration, invalid numbers, and missing files.
- Maintainability – responsibilities are separated into focused Java classes.
- Performance – lightweight file operations and in-memory filtering for a college-scale dataset.
- Resource Efficiency – no external database or server is required.
- Error Handling – invalid input and file errors produce user-facing messages.

6. System Architecture

The application uses a simple layered structure. ExamVaultSystem controls the command-line workflow, ResourceManager handles resource operations, and FileManager handles persistent storage.

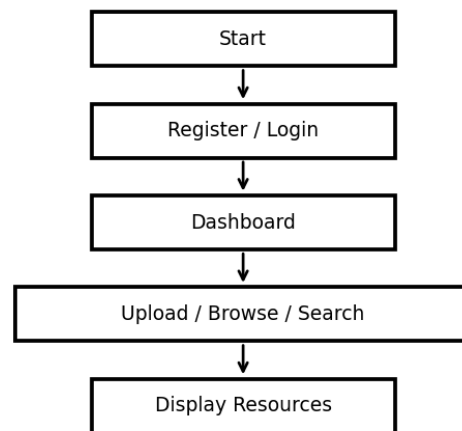
ExamVault System Architecture



7. Process Flow / Workflow

The student starts the application, registers or logs in, uses the dashboard to upload, browse, or search resources, and then logs out or exits.

ExamVault Process Workflow

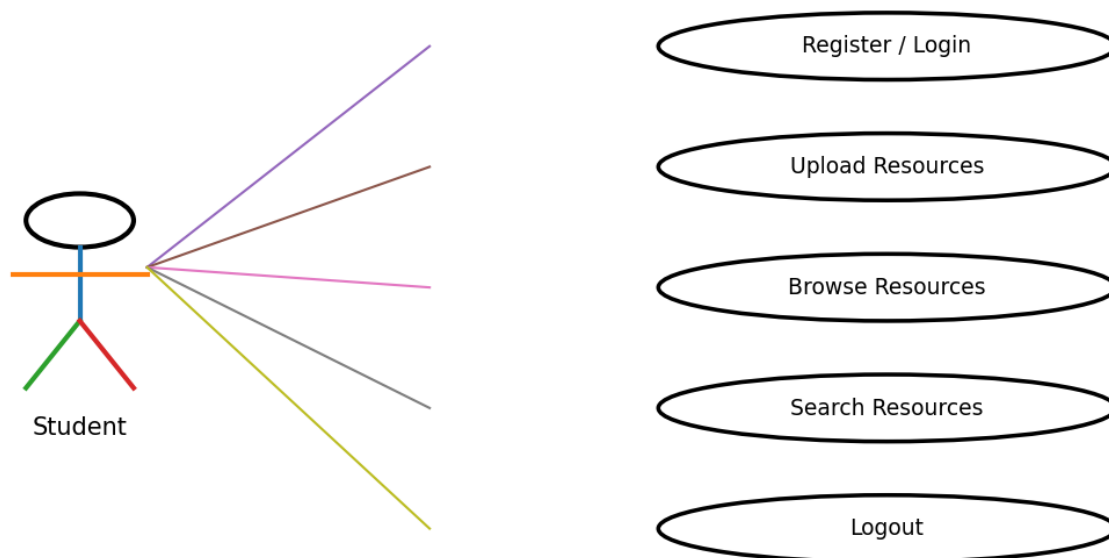


8. Design Diagrams

8.1 UML Use Case Diagram

The primary actor is the Student, who can register/login, upload resources, browse resources, search resources, and log out.

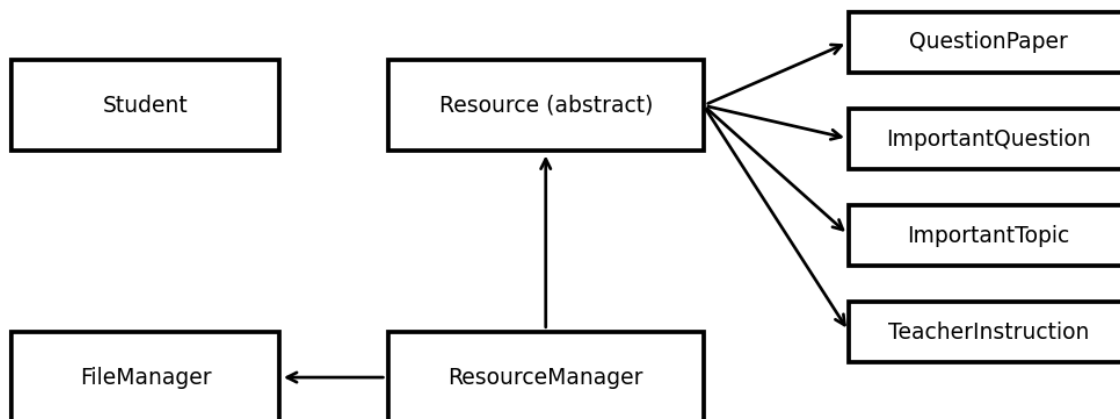
UML Use Case Diagram



8.2 UML Class Diagram

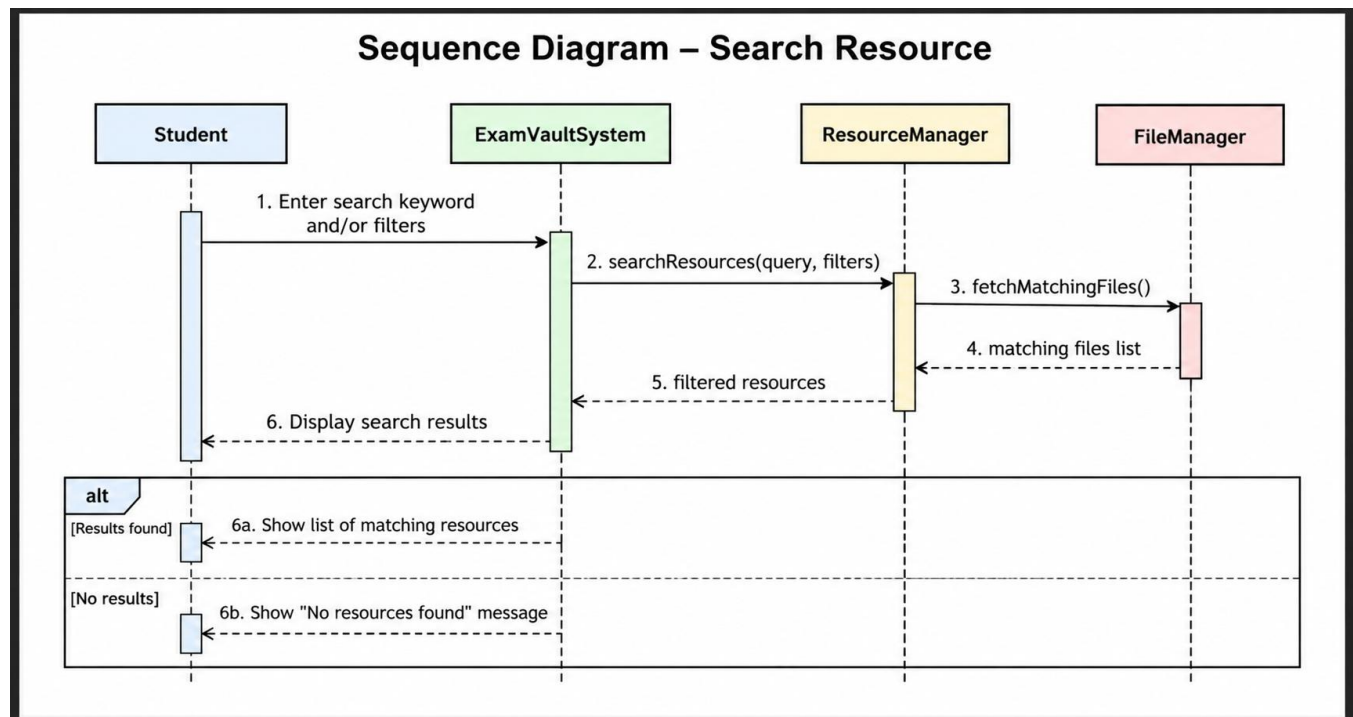
Resource is an abstract base class. QuestionPaper, ImportantQuestion, ImportantTopic, and TeacherInstruction specialize Resource. ResourceManager coordinates resource operations and FileManager provides storage.

UML Class Diagram



8.3 UML Sequence Diagram

This sequence shows a student submitting search filters, the system requesting stored resources, and matching results being displayed.



8.4 Storage Design

Student data is stored in students.txt and resource data is stored in resources.txt. Question paper files can be kept in uploads/.

Storage Design

students.txt				
name	registration	branch	semester	password

resources.txt									
type	subject	exam	slot group	semester	year	title	description	uploader	details

9. Design Decisions & Rationale

Command-Line Interface: Keeps the project fully executable from the command line without external deployment.

Object-Oriented Model: Represents resource categories as subclasses of a common Resource abstraction.

File-Based Persistence: Avoids database setup while demonstrating Java file handling.

ResourceManager: Separates resource management and filtering logic from the interface workflow.

Slot Group: Allows combined groups such as C11/C12/C13 to be treated as one searchable group.

10. Implementation Details

Main – Application entry point.

ExamVaultSystem – Controls registration, login, dashboard, upload, browse, and search workflows.

Student – Represents student account information.

Resource – Abstract base class for common resource attributes.

QuestionPaper – Represents question paper resources.

ImportantQuestion – Represents important-question resources.

ImportantTopic – Represents important-topic resources.

TeacherInstruction – Represents teacher-instruction resources.

ResourceManager – Handles resource creation, loading, browsing, and filtering.

FileManager – Handles persistent file storage.

11. Screenshots / Results

```
=====
                        EXAMVAULT
          College Exam Resource Repository
=====
1. Register
2. Login
3. Exit
Enter choice: █
```

Figure 1 – Application Start / Login

Application Start and Login Interface

This screen displays the initial ExamVault interface, where students can choose to register, log in, or exit the application.

```
=====
                        DASHBOARD
=====
Welcome, Avika Singh
1. Browse All Resources
2. Search Resources
3. Upload Question Paper
4. Upload Important Questions
5. Upload Important Topics
6. Upload Teacher Instruction
7. My Uploads
8. Delete My Upload
9. View Profile
10. Logout
Enter choice: █
```

Figure 2 – Dashboard

Student Dashboard

This screen displays the ExamVault dashboard after successful login, providing access to resource browsing, searching, uploading, profile management, and logout.

```
--- Upload Important Topics ---
Subject: Java
Exam (CAT-1/CAT-2/TEE): CAT-1

Available Slot Groups:
1. A11/A12/A13
2. B11/B12/B13
3. C11/C12/C13
4. D11/D12/D13
5. E11/E12/E13
6. F11/F12/F13
Select slot group: 3
Semester: 3
Academic Year: 2026
Title: Important Java Topics
Description (optional): Topics given for CAT-1
Important topics: Inheritance, Interfaces, Exception Handling
Important Topics uploaded successfully.
```

Figure 3 – Important Topics Upload

Important Topics Upload

This screen displays the upload of important Java topics for CAT-1, including the subject, examination, slot group, semester, academic year, title, and topic details.

```
--- Upload Question Paper ---  
Subject: Java  
Exam (CAT-1/CAT-2/TEE): CAT-1  
  
Available Slot Groups:  
1. A11/A12/A13  
2. B11/B12/B13  
3. C11/C12/C13  
4. D11/D12/D13  
5. E11/E12/E13  
6. F11/F12/F13  
Select slot group: 3  
Semester: 3  
Academic Year: 2026  
Title: Java CAT-1 Question Paper  
Description (optional): Test question paper  
Full path of question paper file: c:\Users\Lenovo\OneDrive\Documents\Java_CAT1_Test.txt  
Question paper uploaded successfully.
```

Figure 4 – Question Paper Upload

Question Paper Upload

This screen displays the process of uploading a Java CAT-1 question paper along with its academic details and file path.

```
===== ALL RESOURCES =====

[1] Important Topics
Title      : Important Java Topics
Subject    : Java
Exam       : CAT-1
Slot Group : A11/A12/A13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Topics     : Inheritance, Interfaces, Exception Handling

[2] Important Topics
Title      : Important Java Topics
Subject    : Java
Exam       : CAT-1
Slot Group : B11/B12/B13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Details    : Topics given for CAT-1
Topics     : Inheritance, Interfaces, Exception Handling

[3] Important Topics
Title      : Java CAT-1 Question Paper
Subject    : Java
Exam       : CAT-1
Slot Group : C11/C12/C13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Topics     : Inheritance, Interfaces, Exception Handling
```

Figure 5 – Browse All Resources

Resource Browsing

This screen displays the resources available in ExamVault, allowing students to view uploaded examination materials organized by their academic details.

```

===== SEARCH RESULTS =====

[1] Important Topics
Title      : Important Java Topics
Subject    : Java
Exam       : CAT-1
Slot Group : A11/A12/A13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Topics     : Inheritance, Interfaces, Exception Handling

[2] Important Topics
Title      : Important Java Topics
Subject    : Java
Exam       : CAT-1
Slot Group : B11/B12/B13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Details    : Topics given for CAT-1
Topics     : Inheritance, Interfaces, Exception Handling

[3] Important Topics
Title      : Java CAT-1 Question Paper
Subject    : Java
Exam       : CAT-1
Slot Group : C11/C12/C13
Semester   : 3
Year       : 2026
Uploaded By : 25BAU10647
Topics     : Inheritance, Interfaces, Exception Handling

```

Figure 6 – Search Results

Resource Search and Filtering

This screen displays the search results generated using selected filters such as subject, examination, slot group, and keyword.

12. Testing Approach

Test	Expected Result	Status
Registration and Login	Registration/login and invalid-	Passed

	login handling work correctly.	
Resource Upload	Important questions, topics, and instructions are saved.	Passed
Question Paper Upload	Valid file path is accepted and stored.	Passed
Browse Resources	Stored resources are displayed.	Passed
Search and Filtering	Subject, exam, slot group, and keyword filters work.	Passed
Invalid File Path	Missing question paper file is rejected.	Passed

13. Challenges Faced

- Designing a reusable resource model for multiple resource categories.
- Maintaining persistent data using files without an external database.
- Validating command-line input and handling invalid file paths.
- Maintaining a modular folder and class structure.

14. Learnings & Key Takeaways

- Practical use of classes, inheritance, abstraction, encapsulation, and polymorphism.
- Java file handling and persistent storage.
- Menu-driven command-line application design.
- Separation of interface, resource management, and storage responsibilities.
- Using GitHub for project version control and documentation.

15. Future Enhancements

- Web-based interface
- Database integration
- Cloud storage
- File preview and downloads
- Advanced search and filtering
- Mobile application support
- Resource ratings and feedback
- Notification system

16. Conclusion

ExamVault provides a centralized and structured approach to sharing college examination resources. It combines student authentication, resource uploading, file-based persistence, browsing, and search functionality in a command-line Java environment. The project demonstrates practical Java Object-Oriented Programming concepts and provides a foundation for future web, mobile, or database-backed extensions.

17. References

1. Oracle Java Documentation – <https://docs.oracle.com/en/java/>
2. Java Platform, Standard Edition Documentation – <https://docs.oracle.com/javase/>
3. GitHub Documentation – <https://docs.github.com/>
4. VIT Bhopal University – VITyarthi: Build Your Own Project, General Project Instructions & Submission Guidelines.